

# K-Modes: Clustering Categorical Data

---

- ❑ K-Means cannot handle non-numerical (categorical) data
  - ❑ Mapping categorical value to 1/0 cannot generate quality clusters for high-dimensional data
- ❑ K-Modes: An extension to K-Means by replacing means of clusters with modes
- ❑ Dissimilarity measure between object  $X$  and the center of a cluster  $Z$ 
  - ❑  $\Phi(x_j, z_j) = 1$  when  $x_j \neq z_j$ ; and 0 otherwise
- ❑ This dissimilarity measure (distance function) is frequency-based
- ❑ Algorithm is still based on iterative object cluster assignment and centroid update
- ❑ A fuzzy K-Modes method is proposed to calculate a fuzzy cluster membership value for each object to each cluster
- ❑ A mixture of categorical and numerical data: Using a K-Prototype method